

SOFTWARE TEST PLAN (STP)

Vehicle Parking System

Document Item	Details
Project	Vehicle Parking System (VPS)
Version	1.0
Authors	Project Team – Team 10
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Related Documents	SRS v1.0; SAD v1.0

1. Introduction

1.1 Purpose

This Software Test Plan defines the scope, strategy, environment, responsibilities, test data, schedule, traceability, metrics, and acceptance criteria for testing the Vehicle Parking System (VPS). The plan converts the requirements documented in the VPS SRS into a controlled verification and validation process and aligns testing with the architecture and interfaces documented in the VPS SAD.

1.2 Scope

Testing covers the software functions for vehicle/user management, parking-slot management, vehicle entry and exit, ticket and receipt generation, fee calculation, payment-status handling, reservations, administrative operations, reporting, security, performance, availability, audit logging, and accessibility.

The following are outside the direct system test boundary:

- Internal implementation of third-party payment gateway services.
- Internal ALPR recognition algorithms supplied by a vendor.
- Physical barrier-controller firmware and vendor hardware internals.
- City-level traffic-management systems and infrastructure outside VPS.

1.3 References

- Vehicle Parking System – Software Requirements Specification (SRS), Version 1.0.
- Vehicle Parking System – Software Architecture Document (SAD), Version 1.0.
- Project database/API design and implementation documentation.
- Requirements Traceability Matrix (RTM).

1.4 Definitions and Acronyms

Term	Definition
VPS	Vehicle Parking System
SRS	Software Requirements Specification
SAD	Software Architecture Document
STP	Software Test Plan
RTM	Requirements Traceability Matrix
ALPR	Automatic License Plate Recognition
UAT	User Acceptance Testing
RBAC	Role-Based Access Control

2. Test Items

Test Item	Main Coverage
User & Authentication Module	Registration/login, logout, roles, authorization, failed-login handling
Vehicle Management Module	Vehicle CRUD, validation, duplicate prevention, lookup
Parking Slot Module	Slot status, availability, allocation, release, categories
Entry Module	Plate capture/manual entry, ticket creation, timestamp, slot assignment, entry barrier
Exit & Fee Module	Ticket validation, parking duration, fee calculation, exit processing
Payment Module	Cash/card/digital-wallet status handling and payment integration
Receipt Module	Receipt generation and delivery/printing

Reservation Module	Advance reservation, validation, expiry and automatic release
Admin Module	Dashboard, occupancy, revenue, history, rate configuration, reports
Database	Integrity, relationships, persistence, transaction consistency, audit data
External Integrations	ALPR, barrier controller, payment gateway interfaces/stubs/sandbox

3. Features to be Tested

Req ID	Feature to Verify	Area
VPS-F-001	License-plate capture through ALPR or manual entry	Entry
VPS-F-002	Assign an available parking slot and mark it occupied	Parking
VPS-F-003	Generate unique parking ticket with entry details	Entry
VPS-F-004	Display real-time slot availability	Parking
VPS-F-005	Support regular, disabled and EV slot categories	Parking
VPS-F-006	Block entry when parking capacity is full	Entry/Parking
VPS-F-007	Calculate fee from entry/exit timestamps and configured rates	Fee
VPS-F-008	Accept and record cash/card/digital-wallet payment status	Payment
VPS-F-009	Generate/print digital or physical receipt	Receipt
VPS-F-010	Release slot when vehicle exits	Exit/Parking
VPS-F-011	Create advance parking reservation	Reservation
VPS-F-012	Automatically release expired reservations	Reservation
VPS-F-013	Admin dashboard for occupancy/revenue/history	Admin
VPS-F-014	Configure parking-rate schedules	Admin
VPS-F-015	Export CSV/PDF operational reports	Admin

3.1 Non-Functional and Security Features

ID	Verification Target	Primary Test
VPS-NF-001	90th-percentile transaction response $\leq$ 5 seconds under normal load	Performance test
VPS-NF-002	99.9% monthly availability excluding scheduled maintenance	Availability/operational test
VPS-NF-003	PCI-DSS-aligned handling; no plaintext card numbers	Security/data inspection
VPS-NF-004	Logs/audit records retained for minimum 5 years	Configuration/data-retention test
VPS-NF-005	WCAG 2.1 AA accessibility target	Accessibility/usability test
VPS-SR-001	TLS 1.2+ for network communication	Security test
VPS-SR-002	Password complexity requirements	Security test
VPS-SR-003	RBAC prevents unauthorized role actions	Authorization test
VPS-SR-004	Lock after 5 failed logins and audit the event	Authentication/security test
VPS-SR-005	License-plate/transaction records protected at rest	Data-protection test

4. Features Not to be Tested

- Internal payment-processing logic, fraud controls, and infrastructure operated by the external payment provider.
- Internal ALPR recognition/model accuracy beyond validating the VPS integration and returned plate data.
- Physical barrier-controller firmware; VPS tests the interface command and response.
- External SMS/email provider internals; VPS verifies request/response handling where applicable.
- City-wide traffic management and unrelated external systems.

These exclusions do not prevent integration, failure-handling, timeout, retry, or mock/sandbox tests of the external interfaces.

5. Test Approach / Strategy

5.1 Test Levels

Level	Purpose	Examples
Unit	Verify individual functions/classes/services	Fee calculation, slot allocation, validation
Integration	Verify interactions between modules	API–DB, payment service–gateway, entry–slot manager
System	Verify complete VPS workflows	Entry-to-exit lifecycle, reservation lifecycle
Acceptance/UAT	Verify SRS acceptance criteria and user workflows	Driver, attendant and admin scenarios

5.2 Test Types

- Functional testing – verify each functional requirement and expected business behavior.

- Regression testing – rerun affected tests after defect fixes or integration changes.
- Performance testing – verify response-time and load-related requirements.
- Security testing – authentication, authorization, input validation, TLS, data protection and audit behavior.
- Usability/accessibility testing – interface clarity, keyboard navigation, contrast, focus and WCAG-oriented checks.
- Database testing – constraints, relationships, persistence, consistency and transaction behavior.
- API testing – request/response validation, status codes, invalid inputs and error handling.

5.3 Entry Criteria

- A testable build is available and deployable in the test environment.
- Required database schema and seed/test data are available.
- Required APIs and dependent modules needed for the planned test cycle are available.
- Test cases for the cycle are reviewed and approved.
- Test environment and required sandbox/mock integrations are operational.

5.4 Exit Criteria

- 100% of planned tests for the release/cycle have been executed, unless formally deferred.
- All critical/blocking defects are closed or formally accepted with documented rationale.
- All high-priority SRS acceptance criteria are verified.
- Required NFR/security tests have been executed and evaluated.
- RTM is updated with implementation and test status.

5.5 Security Validation

- Verify TLS 1.2+ on applicable network connections.
- Verify password policy and RBAC enforcement.
- Verify lockout/audit behavior after five failed login attempts.
- Check that card numbers and other sensitive values are not written to application logs.
- Validate protected storage of license-plate and transaction records.
- Exercise invalid/malicious input paths, including SQL-injection-style and authorization-bypass cases in a controlled test environment.

6. Test Environment

Category	Planned Environment
Frontend	React.js web application
Backend	Java Spring Boot REST services
Database	MySQL 8.x
API Testing	Postman or equivalent API client
Browser	Google Chrome; additional browser checks where required
Performance	JMeter or equivalent load/performance tool, if used by team
Version Control	Git and GitHub
Operating System	Development/test machines running Windows and/or Linux
External Integrations	Payment sandbox/mock, ALPR mock/interface, barrier-controller simulation/stub where hardware is unavailable

6.1 Test Data

- Valid and invalid user credentials for each supported role.
- Valid, invalid and duplicate vehicle registration numbers.
- Available, occupied, reserved, EV and accessibility parking slots.
- Parking tickets with valid, expired, missing and duplicate identifiers.
- Parking durations covering short, normal and long stays.
- Successful, failed, cancelled and unavailable payment responses.
- Reservations that are valid, expired, cancelled and conflicting.
- Rate schedules covering different time periods/categories.

7. Test Schedule

Milestone	Planned Activity	Status
STP Preparation	Finalize test scope, strategy and environment	Planned
Test Case Design	Create functional, NFR and security test cases	Planned
Environment Setup	Configure application, DB, APIs and test data	Planned
Unit Testing	Developer-level module verification	Planned

Integration Testing	Verify module/API/database interactions	Planned
System Testing	Execute end-to-end VPS scenarios	Planned
Security & Performance	Execute security and performance suites	Planned
UAT	Validate acceptance criteria with representative users	Planned
Regression	Retest after defect fixes and final integration	Planned
Test Summary	Compile results, defects, metrics and final RTM	Planned

8. Test Deliverables

- Software Test Plan (this document).
- Detailed Test Case Specification.
- Test data and test scripts where applicable.
- Test execution results/logs.
- Defect/issue records.
- Requirements Traceability Matrix.
- Performance and security test evidence.
- Final Test Summary Report.

9. Roles and Responsibilities

Role / Member	Responsibility
Member 1 – User & Vehicle Management	Implement/test user and vehicle functions; contribute SRS-derived test cases; review authentication/vehicle coverage.
Member 2 – Parking Management	Implement/test slot allocation, availability, categories and reservation behavior; contribute architecture-related verification.
Member 3 – Transactions & Database	Implement/test entry, exit, fee, payment status, history and database integrity.
Member 4 – Admin & Testing Lead	Coordinate STP, test cases, test execution, defect tracking, RTM and test summary; implement/test admin features.
All Team Members	Review test cases, execute assigned tests, report defects, review fixes and maintain GitHub traceability.

10. Risks and Mitigation

Risk	Impact	Mitigation
Stable build delivered late	Test execution delay	Request incremental smoke builds and test modules as they become available.
Database/environment failure	Tests blocked	Maintain reproducible configuration and backup test data.
Payment gateway unavailable	Payment tests blocked	Use sandbox/mock responses and test failure/timeout handling.
ALPR/barrier hardware unavailable	Hardware integration blocked	Use interface mocks/stubs and test manual plate entry.
Integration conflicts between branches	Build instability	Use feature branches, PR review and frequent integration.
Large number of defects near deadline	Release delay	Prioritize critical/high defects and run risk-based regression.

11. Assumptions & Dependencies

- A stable VPS build will be available for each formal test cycle.
- MySQL and required backend services will be accessible.
- Representative test data can be created without using real payment/card information.
- Payment sandbox/mock services can provide success and failure responses.
- ALPR and barrier interfaces can be simulated when physical hardware is unavailable.
- Team members will push test evidence and defect-related changes through GitHub.
- External service outages are not treated as defects in third-party systems, but VPS failure handling is tested.

12. Suspension & Resumption Criteria

12.1 Suspend Testing When

- The test environment is unavailable or unusable for the planned test cycle.
- A critical defect prevents execution of a major workflow such as login, entry, exit or database access.
- The build is sufficiently unstable that a substantial portion of planned tests cannot execute.
- Required test data or dependent sandbox/mock services are unavailable for critical scenarios.

12.2 Resume Testing When

- Blocking defects are fixed and a new testable build is delivered.
- The environment and database are restored.
- Required test data and dependent services are available.
- A smoke test confirms that the build is stable enough for continued execution.

13. Test Case Management & Traceability

Every test case shall map to at least one SRS requirement where applicable. The RTM will connect the requirement to the SAD design/component, implementation module, test case and final execution result.

Requirement	Example Test Case IDs	Design/Module	Final Status
VPS-F-001	TC-ENTRY-001 to TC-ENTRY-003	Entry / ALPR Adapter	Pending
VPS-F-002	TC-SLOT-001 to TC-SLOT-004	Slot Manager	Pending
VPS-F-007	TC-FEE-001 to TC-FEE-004	Fee Calculator	Pending
VPS-F-008	TC-PAY-001 to TC-PAY-004	Payment Service	Pending
VPS-F-010	TC-EXIT-001 to TC-EXIT-003	Exit / Slot Manager	Pending
VPS-F-013	TC-ADMIN-001 to TC-ADMIN-004	Admin Dashboard	Pending
VPS-NF-001	TC-PERF-001	API/Application Layer	Pending
VPS-NF-005	TC-ACC-001	Web UI	Pending
VPS-SR-003	TC-SEC-003	Authentication/RBAC	Pending

14. Test Metrics & Reporting

Metric	Definition
Test execution %	Executed test cases / planned test cases × 100
Pass %	Passed test cases / executed test cases × 100
Fail %	Failed test cases / executed test cases × 100
Blocked %	Blocked test cases / planned test cases × 100
Requirement coverage	Requirements with mapped/executed tests / total testable requirements × 100
Defect count by severity	Number of open/closed defects grouped by critical/high/medium/low
Defect aging	Time defects remain open before resolution
Performance result	Observed response-time/load metrics against VPS-NF-001

Reporting outputs: test execution status, defect summary, requirement coverage, performance/security findings, and the final Test Summary Report.

15. Approvals

Role	Name	Signature / Date
QA / Testing Lead	Aditya	
Development Lead	Vikas	
Project / Faculty Reviewer		